

EcoHub AI - Intelligent Carbon Footprint Prediction and Sustainability Recommendation Platform

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Project Overview

Climate change, rising carbon emissions, and unsustainable resource consumption have become some of the world's most critical challenges. **EcoHub AI** is an intelligent sustainability platform that leverages **Artificial Intelligence, Machine Learning, and Data Analytics** to help individuals understand, monitor, and reduce their environmental impact.

The platform analyzes real-world lifestyle factors including **transportation habits, electricity consumption, water usage, dietary patterns, waste generation, plastic consumption, renewable energy adoption, internet usage, and household characteristics** to accurately predict an individual's carbon footprint. Beyond prediction, EcoHub AI evaluates sustainability behavior through a **Green Score**, identifies user segments using **Clustering**, classifies emission levels through **Machine Learning**, and generates **AI-powered personalized recommendations** to encourage environmentally responsible decisions.

The project integrates **Machine Learning, Deep Learning, Explainable AI, Interactive Dashboards, and Generative AI** to transform raw environmental data into actionable sustainability insights. By combining intelligent prediction with personalized guidance, EcoHub AI empowers users to adopt greener lifestyles while contributing to global climate action.

Key Features

- AI-Based Carbon Footprint Prediction
- Sustainability & Green Score Analysis
- Personalized AI Sustainability Recommendations
- Interactive Analytics Dashboard
- Lifestyle-Based Environmental Impact Assessment
- Machine Learning & Deep Learning Models
- User Behavior Clustering and Classification
- Carbon Emission Trend Analysis
- Renewable Energy Awareness
- Personalized Sustainability Report

Technologies Used

- Python
- Pandas & NumPy
- Matplotlib, Seaborn & Plotly
- Scikit-learn
- TensorFlow / Keras
- Streamlit
- Gemini / OpenAI API
- Git & GitHub

Sustainable Development Goals (SDGs)

This project contributes to multiple United Nations Sustainable Development Goals:

- **SDG 7 – Affordable and Clean Energy**
- **SDG 9 – Industry, Innovation and Infrastructure**
- **SDG 12 – Responsible Consumption and Production**
- **SDG 13 – Climate Action**
- **SDG 17 – Partnerships for the Goals**

Expected Outcome

Develop an intelligent AI-driven sustainability platform capable of predicting carbon footprint, evaluating environmental behavior, generating personalized eco-friendly recommendations, and enabling data-driven decision-making for individuals and organizations to promote sustainable living and reduce carbon emissions.